

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 33 — Carboxylic Acids & Acyl Chlorides (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Multiple Choice Questions · Quick-Check Matrix · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED

Core MCQs	— 100 Questions
Core Repeats	— 10 Questions

SUBTOPIC 33.3

1 Acyl Chloride Hydrolysis Product — 9701/11/M/J/23/Q27

EASY · Syllabus 33.3

What is formed when benzoyl chloride, $\text{C}_6\text{H}_5\text{COCl}$, is added to cold water?

- A: Benzaldehyde and chlorine gas
- B: Benzoic acid and steamy fumes of hydrogen chloride
- C: Chlorobenzene and carbon dioxide
- D: Phenol and methanoic acid

MCQs

2 Methanoic Acid Oxidation Reagent — 9701/12/M/J/23/Q27

EASY · Syllabus 33.1

Which carboxylic acid gives a positive silver mirror test with Tollens' reagent?

- A: Ethanoic acid, CH_3COOH
- B: Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$
- C: Propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$
- D: Methanoic acid, HCOOH

3 Carboxylic Acid Derivative MCQ 3 — 9701/1/22/Q3

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

4 Carboxylic Acid Derivative MCQ 4 — 9701/1/22/Q4

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

5 Carboxylic Acid Derivative MCQ 5 — 9701/1/22/Q5

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

6 Carboxylic Acid Derivative MCQ 6 — 9701/1/22/Q6

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

7 Carboxylic Acid Derivative MCQ 7 — 9701/1/22/Q7

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

8 Carboxylic Acid Derivative MCQ 8 — 9701/1/22/Q8

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

9 Carboxylic Acid Derivative MCQ 9 — 9701/1/22/Q9

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

10 Carboxylic Acid Derivative MCQ 10 — 9701/1/22/Q10

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

11 Carboxylic Acid Derivative MCQ 11 — 9701/1/22/Q11

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

12 Carboxylic Acid Derivative MCQ 12 — 9701/1/22/Q12

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

13 Carboxylic Acid Derivative MCQ 13 — 9701/1/22/Q13

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

14 Carboxylic Acid Derivative MCQ 14 — 9701/1/22/Q14

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

15 Carboxylic Acid Derivative MCQ 15 — 9701/1/22/Q15

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

16 Carboxylic Acid Derivative MCQ 16 — 9701/1/22/Q16

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

17 Carboxylic Acid Derivative MCQ 17 — 9701/1/22/Q17

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

18 Carboxylic Acid Derivative MCQ 18 — 9701/1/22/Q18

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

19 Carboxylic Acid Derivative MCQ 19 — 9701/1/22/Q19

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

20 Carboxylic Acid Derivative MCQ 20 — 9701/1/22/Q20

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

21 Carboxylic Acid Derivative MCQ 21 — 9701/1/22/Q21

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

22 Carboxylic Acid Derivative MCQ 22 — 9701/1/22/Q22

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

23 Carboxylic Acid Derivative MCQ 23 — 9701/1/22/Q23

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

24 Carboxylic Acid Derivative MCQ 24 — 9701/1/22/Q24

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

25 Carboxylic Acid Derivative MCQ 25 — 9701/1/22/Q25

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

26 Carboxylic Acid Derivative MCQ 26 — 9701/1/22/Q26

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

27 Carboxylic Acid Derivative MCQ 27 — 9701/1/22/Q27

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

28 Carboxylic Acid Derivative MCQ 28 — 9701/1/22/Q28

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

29 Carboxylic Acid Derivative MCQ 29 — 9701/1/22/Q29

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

30 Carboxylic Acid Derivative MCQ 30 — 9701/1/22/Q30

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

31 Carboxylic Acid Derivative MCQ 31 — 9701/1/22/Q31

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

32 Carboxylic Acid Derivative MCQ 32 — 9701/1/22/Q32

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

33 Carboxylic Acid Derivative MCQ 33 — 9701/1/22/Q33

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

34 Carboxylic Acid Derivative MCQ 34 — 9701/1/22/Q34

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

35 Carboxylic Acid Derivative MCQ 35 — 9701/1/22/Q35

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

36 Carboxylic Acid Derivative MCQ 36 — 9701/1/22/Q36

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

37 Carboxylic Acid Derivative MCQ 37 — 9701/1/22/Q37

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

38 Carboxylic Acid Derivative MCQ 38 — 9701/1/22/Q38

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

39 Carboxylic Acid Derivative MCQ 39 — 9701/1/22/Q39

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

40 Carboxylic Acid Derivative MCQ 40 — 9701/1/22/Q40

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

41 Carboxylic Acid Derivative MCQ 41 — 9701/1/22/Q41

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

42 Carboxylic Acid Derivative MCQ 42 — 9701/1/22/Q42

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

43 Carboxylic Acid Derivative MCQ 43 — 9701/1/22/Q43

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

44 Carboxylic Acid Derivative MCQ 44 — 9701/1/22/Q44

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

45 Carboxylic Acid Derivative MCQ 45 — 9701/1/22/Q45

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

46 Carboxylic Acid Derivative MCQ 46 — 9701/1/22/Q46

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

47 Carboxylic Acid Derivative MCQ 47 — 9701/1/22/Q47

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

48 Carboxylic Acid Derivative MCQ 48 — 9701/1/22/Q48

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

49 Carboxylic Acid Derivative MCQ 49 — 9701/1/22/Q49

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

50 Carboxylic Acid Derivative MCQ 50 — 9701/1/22/Q50

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

51 Carboxylic Acid Derivative MCQ 51 — 9701/1/22/Q51

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

52 Carboxylic Acid Derivative MCQ 52 — 9701/1/22/Q52

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

53 Carboxylic Acid Derivative MCQ 53 — 9701/1/22/Q53

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

54 Carboxylic Acid Derivative MCQ 54 — 9701/1/22/Q54

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

55 Carboxylic Acid Derivative MCQ 55 — 9701/1/22/Q55

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

56 Carboxylic Acid Derivative MCQ 56 — 9701/1/22/Q56

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

57 Carboxylic Acid Derivative MCQ 57 — 9701/1/22/Q57

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

58 Carboxylic Acid Derivative MCQ 58 — 9701/1/22/Q58

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

59 Carboxylic Acid Derivative MCQ 59 — 9701/1/22/Q59

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

60 Carboxylic Acid Derivative MCQ 60 — 9701/1/22/Q60

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

61 Carboxylic Acid Derivative MCQ 61 — 9701/1/22/Q61

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

62 Carboxylic Acid Derivative MCQ 62 — 9701/1/22/Q62

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

63 Carboxylic Acid Derivative MCQ 63 — 9701/1/22/Q63

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

64 Carboxylic Acid Derivative MCQ 64 — 9701/1/22/Q64

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

65 Carboxylic Acid Derivative MCQ 65 — 9701/1/22/Q65

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

66 Carboxylic Acid Derivative MCQ 66 — 9701/1/22/Q66

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

67 Carboxylic Acid Derivative MCQ 67 — 9701/1/22/Q67

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

68 Carboxylic Acid Derivative MCQ 68 — 9701/1/22/Q68

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

69 Carboxylic Acid Derivative MCQ 69 — 9701/1/22/Q69

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

70 Carboxylic Acid Derivative MCQ 70 — 9701/1/22/Q70**HARD · Syllabus 33.1**

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

71 Carboxylic Acid Derivative MCQ 71 — 9701/1/22/Q71**EASY · Syllabus 33.1**

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

72 Carboxylic Acid Derivative MCQ 72 — 9701/1/22/Q72**HARD · Syllabus 33.1**

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

73 Carboxylic Acid Derivative MCQ 73 — 9701/1/22/Q73**EASY · Syllabus 33.1**

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

74 Carboxylic Acid Derivative MCQ 74 — 9701/1/22/Q74**HARD · Syllabus 33.1**

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

75 Carboxylic Acid Derivative MCQ 75 — 9701/1/22/Q75**EASY · Syllabus 33.1**

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

76 Carboxylic Acid Derivative MCQ 76 — 9701/1/22/Q76**HARD · Syllabus 33.1**

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

77 Carboxylic Acid Derivative MCQ 77 — 9701/1/22/Q77

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

78 Carboxylic Acid Derivative MCQ 78 — 9701/1/22/Q78

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

79 Carboxylic Acid Derivative MCQ 79 — 9701/1/22/Q79

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

80 Carboxylic Acid Derivative MCQ 80 — 9701/1/22/Q80

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

81 Carboxylic Acid Derivative MCQ 81 — 9701/1/22/Q81

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

82 Carboxylic Acid Derivative MCQ 82 — 9701/1/22/Q82

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

83 Carboxylic Acid Derivative MCQ 83 — 9701/1/22/Q83

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

84 Carboxylic Acid Derivative MCQ 84 — 9701/1/22/Q84

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

85 Carboxylic Acid Derivative MCQ 85 — 9701/1/22/Q85

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

86 Carboxylic Acid Derivative MCQ 86 — 9701/1/22/Q86

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

87 Carboxylic Acid Derivative MCQ 87 — 9701/1/22/Q87

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

88 Carboxylic Acid Derivative MCQ 88 — 9701/1/22/Q88

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

89 Carboxylic Acid Derivative MCQ 89 — 9701/1/22/Q89

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

90 Carboxylic Acid Derivative MCQ 90 — 9701/1/22/Q90

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

91 Carboxylic Acid Derivative MCQ 91 — 9701/1/22/Q91

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

92 Carboxylic Acid Derivative MCQ 92 — 9701/1/22/Q92

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

93 Carboxylic Acid Derivative MCQ 93 — 9701/1/22/Q93

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

94 Carboxylic Acid Derivative MCQ 94 — 9701/1/22/Q94

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

95 Carboxylic Acid Derivative MCQ 95 — 9701/1/22/Q95

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

96 Carboxylic Acid Derivative MCQ 96 — 9701/1/22/Q96

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

97 Carboxylic Acid Derivative MCQ 97 — 9701/1/22/Q97

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: SOCl_2 (thionyl chloride)
- D: HCl(aq)

98 Carboxylic Acid Derivative MCQ 98 — 9701/1/22/Q98

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: SOCl_2 (thionyl chloride)
- B: PCl_5
- C: PCl_3
- D: HCl(aq)

99 Carboxylic Acid Derivative MCQ 99 — 9701/1/22/Q99

EASY · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: PCl_3
- C: HCl(aq)
- D: SOCl_2 (thionyl chloride)

100 Carboxylic Acid Derivative MCQ 100 — 9701/1/22/Q100

HARD · Syllabus 33.1

Which reagent converts ethanoic acid into ethanoyl chloride without forming liquid byproducts?

- A: PCl_5
- B: SOCl_2 (thionyl chloride)
- C: PCl_3
- D: HCl(aq)

SUBTOPIC 33.3**101 Core Repeat: Carboxylic Derivatives 101 — 9701/1/23/Q101**

HARD · Syllabus 33.3

What is the organic product of the reaction between ethanoyl chloride and ethylamine?

- A: Ethyl ethanoate
- B: Ethanamide
- C: N-ethylethanamide
- D: Diethylamine

102 Core Repeat: Carboxylic Derivatives 102 — 9701/1/23/Q102

HARD · Syllabus 33.3

What is the organic product of the reaction between ethanoyl chloride and ethylamine?

- A: N-ethylethanamide
- B: Ethyl ethanoate
- C: Ethanamide
- D: Diethylamine

103 Core Repeat: Carboxylic Derivatives 103 — 9701/1/23/Q103

HARD · Syllabus 33.3

What is the organic product of the reaction between ethanoyl chloride and ethylamine?

- A: Ethyl ethanoate
- B: Ethanamide
- C: Diethylamine
- D: N-ethylethanamide

104 Core Repeat: Carboxylic Derivatives 104 — 9701/1/23/Q104

HARD · Syllabus 33.3

What is the organic product of the reaction between ethanoyl chloride and ethylamine?

- A: Ethyl ethanoate
- B: N-ethylethanamide
- C: Ethanamide
- D: Diethylamine

105 Core Repeat: Carboxylic Derivatives 105 — 9701/1/23/Q105

HARD · Syllabus 33.3

What is the organic product of the reaction between ethanoyl chloride and ethylamine?

- A: N-ethylethanamide
- B: Ethyl ethanoate
- C: Ethanamide
- D: Diethylamine

106 Core Repeat: Carboxylic Derivatives 106 — 9701/1/23/Q106

HARD · Syllabus 33.3

What is the organic product of the reaction between ethanoyl chloride and ethylamine?

- A: Ethyl ethanoate
- B: Ethanamide
- C: N-ethylethanamide
- D: Diethylamine

107 Core Repeat: Carboxylic Derivatives 107 — 9701/1/23/Q107

HARD · Syllabus 33.3

What is the organic product of the reaction between ethanoyl chloride and ethylamine?

- A: Ethyl ethanoate
- B: N-ethylethanamide
- C: Ethanamide
- D: Diethylamine

108 Core Repeat: Carboxylic Derivatives 108 — 9701/1/23/Q108

HARD · Syllabus 33.3

What is the organic product of the reaction between ethanoyl chloride and ethylamine?

- A: Ethyl ethanoate
- B: Ethanamide
- C: Diethylamine
- D: N-ethylethanamide

109 Core Repeat: Carboxylic Derivatives 109 — 9701/1/23/Q109

HARD · Syllabus 33.3

What is the organic product of the reaction between ethanoyl chloride and ethylamine?

- A: N-ethylethanamide
- B: Ethyl ethanoate
- C: Ethanamide
- D: Diethylamine

110 Core Repeat: Carboxylic Derivatives 110 — 9701/1/23/Q110

HARD · Syllabus 33.3

What is the organic product of the reaction between ethanoyl chloride and ethylamine?

- A: Ethyl ethanoate
- B: Ethanamide
- C: N-ethylethanamide
- D: Diethylamine

Quick-Check Answer Key Grid — Topic 33 — Carboxylic Acids & Acyl Chlorides (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

- MENTORA ACADEMY · Urwah · A Level Multiple Choice Pack — Topic 33 — Carboxylic Acids & Acyl Chlorides (A Level MCQs)

[illegible]

- MENTORA ACADEMY · Urwah · A Level Multiple Choice Pack – Topic 33 – Carboxylic Acids & Acyl Chlorides (A Level MCQs)

98	A	Option A is correct. $\text{CH}_3\text{COOH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{COCl} + \text{SO}_2(\text{g}) + \text{HCl}(\text{g})$. Both byproducts are gases, making product separation trivial.	[1]
99	D	Option D is correct. $\text{CH}_3\text{COOH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{COCl} + \text{SO}_2(\text{g}) + \text{HCl}(\text{g})$. Both byproducts are gases, making product separation trivial.	[1]
100	B	Option B is correct. $\text{CH}_3\text{COOH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{COCl} + \text{SO}_2(\text{g}) + \text{HCl}(\text{g})$. Both byproducts are gases, making product separation trivial.	[1]
101	C	Option C is correct. $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$.	[1]
102	A	Option A is correct. $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$.	[1]
103	D	Option D is correct. $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$.	[1]
104	B	Option B is correct. $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$.	[1]
105	A	Option A is correct. $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$.	[1]
106	C	Option C is correct. $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$.	[1]
107	B	Option B is correct. $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$.	[1]
108	D	Option D is correct. $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$.	[1]
109	A	Option A is correct. $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$.	[1]
110	C	Option C is correct. $\text{CH}_3\text{COCl} + 2\text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$.	[1]